Institutions for Distributed Discovery

Evidence Acquisition, Adaptation, Coverage, Incentives, and Measurement

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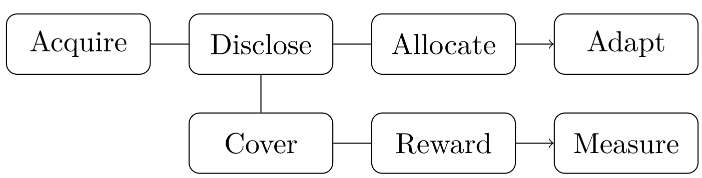
Abstract

Distributed discovery turns evidence into a portfolio of actions. This working paper organizes completed bounded results by an institutional stack: acquire, disclose, allocate, adapt, cover, reward, and measure. The taxonomy is an organizing device, not an empirical theorem. Every numerical statement is tied to a study claim and immutable run in the public repository.

Evidence status. DD-004–DD-006A are exact bounded computations; DD-007 is synthetic-only; DD-008 is an exact synthetic source-choice fixture. No real-institutional claim or peer-review claim is made.

1 The Discovery Stack



The stack separates margins that are often conflated. Acquisition determines which signals exist; disclosure determines what becomes shared; allocation maps information to actions; adaptation uses feedback; coverage evaluates union value; rewards alter private incentives; measurement determines what can be inferred.

2 Evidence discipline

The stack is deliberately not a single optimization problem. A result at one margin can be useful while leaving other margins unspecified. The repository therefore records a claim type, scope statement, immutable manifest, and output checksums for each finding; aggregation in this paper does not upgrade a claim.

3 Acquisition

DD-008 compares a free common signal with a costly independent signal in a two-agent, three-target direct-clue-following fixture. With signal accuracy p and independent-source cost c, common/common has social value p; one independent source has $2p - p^2 - c$. Private source choice

Table 1: Evidence inventory for the Discovery Stack. Each row is a bounded registered result, not a universal institutional law.

Margin	Registered object	Evidence	Claim
Adapt	16 schedule records under perfect elimination	exact fixture	DD-C-0045
Cover	3 coverage-frontier records	bounded witnesses	DD-C-0046, DD-C-0047
Reward	123 normalized-transfer rows	exhaustive class	DD-C-0050
Acquire	5 common-source trap cells	exact source-choice grid	DD-C-0051
Measure	synthetic event generator and recovery checks	synthetic-only	DD-C-0049

Generator: `distributed_discovery.papers.build_discovery_institutions`; inputs are checksum-validated from immutable manifests.

can retain common/common when one independent source has higher net social value. This is a bounded accounting result, not a general information-public-good theorem. The diagnosis is useful because a disappointing portfolio may reflect source correlation, poor sharing, or convergent actions—three distinct interventions.

4 Disclosure and allocation

DD-001 and DD-002 show that sharing evidence and selecting an action are distinct operations. Role policies can improve a finite private-information fixture, and the effect of a disclosure refinement depends on the declared equilibrium selection rule. These are claim-specific bounded results rather than a monotonicity failure for all information structures. Accordingly, planner, pure-equilibrium, and selected-equilibrium quantities are reported separately rather than collapsed into a claim that "information helps."

5 Adaptation

In DD-004's perfect-elimination fixture, batch schedules share terminal discovery but differ in expected actions and rounds. Thus terminal discovery and operating cost are distinct objectives; no noisy-test or general adaptivity conclusion follows. Terminal discovery and operating cost are thus distinct objectives. A general sequential design conclusion would require noisy outcomes and a stopping model.

6 Coverage

DD-005's finite overlap witnesses show why action labels and individual rankings are insufficient summaries of portfolio value. The results do not contradict general approximation guarantees for monotone submodular objectives. The relevant object is a portfolio, not a list of individually high-scoring actions; this remains a finite modeling warning, not a universal ranking result.

7 Rewards

DD-006 separates weak from strict implementation. The score-difference census and the broader normalized linear frontier both contain weak rows but no strict truthful/direct implementation under the registered observability and tie rules. This is not dominant-strategy implementation or an arbitrary-transfer-table impossibility result. The normalized linear census leaves nonlinear, contingent, and differently observable mechanisms open for new registrations.

8 Measurement

DD-007 validates synthetic event records and estimator recovery only under its generator. Matching error creates attenuation and explicit counterexamples show that action duplication or missing provenance alone need not identify copying or effective channels. Measurement is part of the institution: without source and action provenance, multiple behavioral mechanisms can fit the same aggregate record.

9 Institutional implications and limitations

The program supports a design discipline: name the evidence source, the action portfolio, feedback process, reward observability, and measurement target before claiming improvement. The studies are small, deliberately registered fixtures. They do not identify universal institutional laws, analyze real data, or establish causal effects in organizations. The synthesis recommends a sequence: name evidence, sharing, allocation, diversification, feedback, feasible rewards, and measurable outcomes before claiming an institutional improvement.

10 Research agenda

Future work requires new registrations for noisy adaptation, stochastic coverage, arbitrary transfer tables, richer source-choice games, and any real-data protocol.

References